DOTE 6635: Artificial Intelligence for Business Research (Spring 2026)

What's Next in DOTE 6635?

Renyu (Philip) Zhang

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Session	Date	Topic	Key Words
1	1.06	AI/ML in a Nutshell	Course Intro, Prediction in Biz Research, ML Model Evaluations
2	1.13	Intro to DL	DL Intro, Neural Nets, DL Computations
3	1.20	RL (I)	Tabular MDP, Bellman Equation
4	1.27	RL (II)	Model-free Prediction & Control, Monte Carlo, Temporal Difference
5	2.03	RL (III)	Value-based DRL, DQN, AlphaGo
6	2.10	RL (IV)	Policy-based DRL, Policy Gradient, PPO, A3C
7	2.24	RL (V)	Off-Policy Evaluation, Off-Policy Learning
8	3.3	RL (VI)	RLHF, RL with Verifiable Reward, LLM Reasoning, GRPO
9	3.10	Agentic AI (I)	Building LLM Agents, ReAct, ToT, MCP
10	3.17	Agentic AI (II)	Agent Training, Tool Use, Planning, Coding Agent
11	3.24	Agentic AI (III)	Agent Evaluation, Benchmarks
12	3.31	Agentic AI (IV)	Multi-Agent Systems, Agentic Simulations
13	4.14	Agentic AI (V)	Business Applications of Agentic AI
13+	2027	Season 4	More to come...

Note Scribing

- Please sign up here: https://docs.google.com/spreadsheets/d/1YIwCR-X8VVLTv-OfGr7DqZJF6YIll-JE3SGI1q7JEus/edit?gid=1567564712#gid=1567564712
- Please make sure every week has some scribers.
- I have prepared some AI-generated notes for your reference: https://github.com/rphilipzhang/AI-PhD-S26/tree/main/Notes/AI-Generated
- If multiple groups sign up to scribe the notes together for one week, please coordinate to prepare one Overleaf project.
- If one topic (i.e., 1 slides deck) is covered by multiple groups of different weeks, please coordinate to prepare one Overleaf project.
- Overleaf template can be found here: https://www.overleaf.com/read/bmfygjgsyzrr#a72daa

Jan 13, 2026

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Session	Date	Paper
1	Jan/6	No Presentation
2	Jan/13	No Presentation
3	Jan/20	Peng, T., Gui, G., Merlau, D. J., Fan, G. J., Sliman, M. B., Brucks, M., ... & Toubia, O. (2025). A mega-study of digital twins reveals strengths, weaknesses and opportunities for further improvement. arXiv preprint arXiv:2509.19088.
4	Jan/27	Ludwig, J., Mullainathan, S., & Rambachan, A. (2025). Large language models: An applied econometric framework (No. w33344). National Bureau of Economic Research.
5	Feb/3	de Kok, T. (2025). ChatGPT for textual analysis? How to use generative LLMs in accounting research. Management Science.
6	Feb/10	Reisenbichler, M., Reutterer, T., & Schweidel, D. A. (2025). Applying large language models to sponsored search advertising. Marketing Science.
7	Feb/24	Burnap, Alex , John R. Hauser, Artem Timoshenko (2023) Product Aesthetic Design: A Machine Learning Augmentation. Marketing Science 42(6):1029-1056.
8	Mar/3	Ye, Z., Yoganarasimhan, H., & Zheng, Y. (2025). Lola: Llm-assisted online learning algorithm for content experiments. Marketing Science.
9	Mar/10	Liu, X. (2023). Dynamic coupon targeting using batch deep reinforcement learning: An application to livestream shopping. Marketing Science, 42(4), 637-658.
10	Mar/17	Chenyu Huang, Zhengyang Tang, Shixi Hu, Ruoqing Jiang, Xin Zheng, Dongdong Ge, Benyou Wang, Zizhuo Wang (2025) ORLM: A Customizable Framework in Training Large Models for Automated Optimization Modeling. Operations Research 73(6):2986-3009.
11	Mar/24	Manning, B. S., & Horton, J. J. (2025). General social agents. arXiv preprint arXiv:2508.17407.
12	Mar/31	Song, Y., & Sun, T. (2024). Ensemble experiments to optimize interventions along the customer journey: A reinforcement learning approach. Management Science, 70(8), 5115-5130.
13	Apr/14	Yang, Y., Zhang, Y., Wu, M., Zhang, K., Zhang, Y., Yu, H., ... & Wang, B. (2025). TwinMarket: A Scalable Behavioral and Social Simulation for Financial Markets. arXiv preprint arXiv:2502.01506.

Replication Project

- Please sign up here: https://docs.google.com/spreadsheets/d/1YIwCR-X8VVLTv-OfGr7DqZJF6YIll-JE3SGI1q7JEus/edit?usp=sharing
- Please make sure every project has some replicators.
- Each group should create a GitHub repo and write a technical blog.
- If multiple groups sign up to replicate one project, please coordinate to present in one week.
- Every group should prepare a 20-min video for your replication project.

Jan 13, 2026

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Scribed Note and Replication Project

- See this spreadsheet for the confirmed schedule: https://docs.google.com/spreadsheets/d/1YIwCR-X8VVLTv-OfGr7DqZJF6YIll-JE3SGI1q7JEus/edit?usp=sharing
- Scribed notes are due **2 weeks from the lecture**.
- Please directly send the link to your scribed note to Philip via philipzhang@cuhk.edu.hk. Do NOT publicize it for now.
- GitHub repo, technical blog, and video are **2 weeks from the presentation** and should be **open-sourced**.
  - Put your code, data, slides deck, technical blog and the link to your video in the GitHub repo.
- Please also share the GitHub and video links here: https://docs.google.com/spreadsheets/d/1YIwCR-X8VVLTv-OfGr7DqZJF6YIll-JE3SGI1q7JEus/edit?gid=422252848#gid=422252848

Jan 20, 2026

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Replication Presentation

- See this spreadsheet for the confirmed schedule: https://docs.google.com/spreadsheets/d/1YIwCR-X8VVLTv-OfGr7DqZJF6YIll-JE3SGI1q7JEus/edit?gid=422252848#gid=422252848

Ludwig, J., Mullainathan, S., & Rambachan, A. (2025). Large language models: An applied econometric framework (No. w33344). National Bureau of Economic Research.

- Presenters: Dengdeng Lin & Yufan Iris Zhang, from CUHK IS
- 30 mins including Q&A

Jan 27, 2026

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Replication Presentation

- See this spreadsheet for the confirmed schedule: https://docs.google.com/spreadsheets/d/1YIwCR-X8VVLTv-OfGr7DqZJF6YIll-JE3SGI1q7JEus/edit?gid=422252848#gid=422252848

de Kok, T. (2025). ChatGPT for textual analysis? How to use generative LLMs in accounting research. *Management Science.*

- Presenters: Aixin Cui & Haojian Zhang
- 30 mins including Q&A

Feb 3, 2026

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Replication Presentation

- See this spreadsheet for the confirmed schedule: https://docs.google.com/spreadsheets/d/1YIwCR-X8VVLTv-OfGr7DqZJF6YIll-JE3SGI1q7JEus/edit?gid=422252848#gid=422252848

Reisenbichler, M., Reutterer, T., & Schweidel, D. A. (2025). Applying large language models to sponsored search advertising. *Marketing Science*.

- Presenters: Qiuhong Wei & Bozhi Xu
- 30 mins including Q&A

Feb 10, 2026

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Replication Presentation

- See this spreadsheet for the confirmed schedule: https://docs.google.com/spreadsheets/d/1YIwCR-X8VVLTv-OfGr7DqZJF6YIll-JE3SGI1q7JEus/edit?gid=422252848#gid=422252848

Burnap, Alex , John R. Hauser, Artem Timoshenko (2023) Product Aesthetic Design: A Machine Learning Augmentation. *Marketing Science* 42(6):1029-1056.

- Presenters: Runming Huang & Leran Li
- 30 mins including Q&A

Feb 24, 2026

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Replication Presentation

- See this spreadsheet for the confirmed schedule: https://docs.google.com/spreadsheets/d/1YIwCR-X8VVLTv-OfGr7DqZJF6YIll-JE3SGI1q7JEus/edit?gid=422252848#gid=422252848

Ye, Z., Yoganarasimhan, H., & Zheng, Y. (2025). Lola: Llm-assisted online learning algorithm for content experiments. *Marketing Science.*

- Presenters: Yusi Chen & Jinghan Liu
- 30 mins including Q&A
- We will return CYT 209A starting from the next week.

Mar 3, 2026

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Final Project

- You should schedule a meeting (online or offline) with me to discuss about your final project.
  - I suggest you do so before submitting your proposal.
- You need to submit a 1-page proposal by 12:30pm on March 10 (soft deadline), articulating your research problem, the data you plan to collect (or have collected), and the AI method you plan to use.
- You need to complete a (short, no more than 10 pages excluding references and appendices) research paper (with research question, data, implementation and, at least partial, results).
- You will also need to record a video (no more than 15 minutes) of your final project.
- Both the paper and the video will be due in early May.
- You will be judged by:
  - (20%) Research questions and methodologies;
  - (40%) Execution and results;
  - (20%) Presentation;
  - (20%) Writing of final paper.
- The paper should be either empirical or ML-methodological (unless with justifiable reasons).

Mar 3, 2026

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Final Project

- Answer the following questions before working on your chosen project:
  - Why do I think this is an interesting research question to business/econ audience? Will I make methodological or empirical contributions?
  - Why do I think this data set is good enough for me to answer this question? How will my results depend on the outcomes of the data?
  - Am I using the cutting-edge AI/ML technologies to solve this problem? Why or why not?

Mar 3, 2026

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Data Sets

- CS Datasets:
  - Kaggle
  - Packages
  - Papers that publish datasets
- Econ Datasets:
  - FOMC, 10-K, government contracting, census data, and other government datasets
  - Papers that publish datasets
- Business Datasets:
  - Published datasets from companies, such as Tmall, JD.COM, Tweet (X), Yelp and Reddit (most of them are on Kaggle)
  - A lot of UTD journals require publicizing data for replication (JF, RFS, IJoC, JM, MktSci, ManSci, OR, and ASQ).
  - Proprietary datasets
- Other Datasets:
  - Scraped data (leverage AI to find new data sets)
  - Crypto

Mar 3, 2026

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Replication Presentation

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Liu, X. (2023). Dynamic coupon targeting using batch deep reinforcement learning: An application to livestream shopping. *Marketing Science*, 42(4), 637-658.

- Presenters: Luxuan Liu & Zhongyuan Zhou
- 30 mins including Q&A

Mar 10, 2026

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Final Project

At your earliest convenience, please submit to me by email a one-page proposal of your final project, articulating your research problem, the data you plan to collect (or have collected), and the AI method you plan to use.

Mar 10, 2026

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Replication Presentation

- See this spreadsheet for the confirmed schedule: https://docs.google.com/spreadsheets/d/1YIwCR-X8VVLTv-OfGr7DqZJF6YIll-JE3SGI1q7JEus/edit?gid=422252848#gid=422252848

Chenyu Huang, Zhengyang Tang, Shixi Hu, Ruoqing Jiang, Xin Zheng, Dongdong Ge, Benyou Wang, Zizhuo Wang (2025) ORLM: A Customizable Framework in Training Large Models for Automated Optimization Modeling. *Operations Research* 73(6):2986-3009.

- Presenters: Lizhuo Xie & Zewei Yu
- 30 mins including Q&A

Mar 17, 2026

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What's Next?

Date	Session	Lecture Topics	Replication	Homework Due
March 17	10	PPO, RLHF, DPO	ORLM	
March 24	11	CoT, RLVR, GRPO, Reasoning Models	General Social Agents	PS6 (RL w. Stable Baselines)
March 31	12	Off-Policy Evaluation, Off-Policy Learning	RL for Experimentation	
April 14	13	AI Agents, Agentic Simulations, Course Wrap-up	Financial Agentic Simulation	PS8 (RLHF), PS3 (Training GPT)
May 1, 11:59pm	N.A.			PS7 (Agentic Workflow), Scribed Lecture Notes
May 10, 11:59pm	N.A.			Project Paper, Slides, Video, and Code
May 11	N.A.			Grade Submission

- With your permission, the scribed lecture notes, project papers, slides, videos, and code will be shared with the students who take this course for credit.

Mar 17, 2026

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Replication Presentation

- See this spreadsheet for the confirmed schedule: https://docs.google.com/spreadsheets/d/1YIwCR-X8VVLTv-OfGr7DqZJF6YIll-JE3SGI1q7JEus/edit?gid=422252848#gid=422252848

Manning, B. S., & Horton, J. J. (2025). General social agents. *arXiv preprint* arXiv:2508.17407.

- Presenter: Keyu Chen
- 30 mins including Q&A

Mar 24, 2026

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What's Next?

Date	Session	Lecture Topics	Replication	Homework Due
March 24	11	DPO, CoT, RLVR, GRPO, Reasoning Models	General Social Agents	PS6 (RL w. Stable Baselines)
March 31	12	Agentic AI Basics	RL for Experimentation	
April 14	13	Agentic Simulations, Course Wrap-up	Financial Agentic Simulation	PS8 (RLHF); PS3 (Training GPT)
May 1, 11:59pm	N.A.			PS7 (Auto-replication); Scribed Lecture Notes
May 10, 11:59pm	N.A.			Project Paper, Slides, Video, and Code; PS9 (Auto-research)
May 11	N.A.			Grade Submission

- With your permission, the scribed lecture notes, project papers, slides, videos, and code will be shared with the students who take this course for credit.

Mar 24, 2026

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Replication Presentation

- See this spreadsheet for the confirmed schedule: https://docs.google.com/spreadsheets/d/1YIwCR-X8VVLTv-OfGr7DqZJF6YIlI-JE3SGI1q7JEus/edit?gid=422252848#gid=422252848

Song, Y., Sun, T. (2024). Ensemble experiments to optimize interventions along the customer journey: A reinforcement learning approach. *Management Science*, 70(8), 5115-5130.

- Presenter: Mingxia Li & Cheng Peng
- 30 mins including Q&A

Mar 31, 2026

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What's Next?

Date	Session	Lecture Topics	Replication	Homework Due
March 31	12	Reasoning LLM, LLM Agents	RL for Experimentation	
April 14	13	Agentic Simulations, Course Wrap-up	Financial Agentic Simulation	PS8 (RLHF); PS3 (Training GPT)
May 1, 11:59pm	N.A.			PS7 (Auto-replication); Scribed Lecture Notes
May 10, 11:59pm	N.A.			Project Paper, Slides, Video, and Code; PS9 (Auto-research)
May 11	N.A.			Grade Submission

- With your permission, the scribed lecture notes, project papers, slides, videos, and code will be shared with the students who take this course for credit.

Mar 31, 2026

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Replication Presentation

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Yang, Y., et al. (2025). TwinMarket: A Scalable Behavioral and Social Simulation for Financial Markets. *arXiv preprint* arXiv:2502.01506.

- Presenter: Chloe Wang & Dream Yu
- 30 mins including Q&A

Apr 14, 2026

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What's Next?

Date	Session	Lecture Topics	Replication	Homework Due
April 14	13	Agentic AI, Course Wrap-up	Financial Agentic Simulation	PS8 (RLHF); PS3 (Training GPT)
May 1, 11:59pm	N.A.			PS7 (Auto-replication); Scribed Lecture Notes
May 10, 11:59pm	N.A.			Project Paper, Slides, Video, and Code; PS9 (Auto-research)
May 11	N.A.			Grade Submission

- With your permission, the scribed lecture notes, project papers, slides, videos, and code will be shared with the students who take this course for credit.

Apr 14, 2026

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